

Evan Frangipane

☎ +1 (310) 310-9459 frangipane.evan@gmail.com 🏠 Los Angeles, CA

🌐 [Linkedin](#) 🌐 [evanfrang](#) [evanfrangipane.com](#)

Education

Ph.D. in Physics *University of California, Santa Cruz*

Santa Cruz, CA *Aug 2018 - Aug 2024*

B.A. in Physics *University of California, Berkeley*

Berkeley, CA *Aug 2014 - Jun 2018*

Certifications

Google Advanced Data Analytics Professional Certificate

Feb 2025

Employment

Graduate Student Researcher, *University of California, Santa Cruz*

Aug 2018 - Aug 2024

- Designed and executed Python-based data analysis pipelines to extract insights from large, complex datasets.
- Conducted hypothesis testing, causal inference analysis, and statistical modeling to support data-driven decision-making.

Teaching Assistant, *University of California, Santa Cruz*

Aug 2018 - Aug 2024

- Taught Python-based data analysis, covering statistical techniques, regression modeling, and machine learning applications.

Technical Projects

End-to-End arXiv Abstract Classification (Self-Directed)

July 2025

- Developed and deployed a logistic regression model for real-time classification of arXiv abstracts, enabling live prediction serving in production.
- Experimented with fine-tuning DistilBERT using Hugging Face Transformers and LoRA (Low-Rank Adaptation), achieving improved accuracy with parameter-efficient training methods.

Happiness Trends Worldwide: A Comparative View 2015 - 2024 (Self-directed)

Mar 2025

- Created a Tableau dashboard using World Happiness Report datasets with maps charting the change in happiness.

Multi-Model Fraud Detection: LR, SVM, and Random Forest (Self-directed)

Feb 2025

- Compared three models for credit card fraud detection on highly imbalanced (280k transactions, < 0.2% fraud) dataset.
- Random Forest outperformed logistic regression and SVM with an 0.84 AUCPR and 0.82 f1-score.

Multiple Testing Problem and the Bonferroni Correction (Self-directed)

Jan 2025

- Simulated the multiple testing problem using Python to analyze how repeated statistical tests inflate false positive rates.
- Implemented Monte Carlo simulations and applied the Bonferroni correction to mitigate inflated significance levels.

Statistical Analysis of Primordial Black Hole Hypotheses

Oct 2023

- Performed large-scale microlensing simulations using Kubernetes, efficiently running parallelized jobs on a distributed computing cluster to save hundreds of hours
- Applied Anderson-Darling test to differentiate population distributions of Planets and Primordial Black Holes

Skills

- **Programming & Tools:** Python (Pandas, scikit-learn, Pytorch), SQL, Git, Docker, Kubernetes, Tableau, $\LaTeX$
- **Machine Learning & Statistical Analysis:** Supervised & Unsupervised Learning, Regression, A/B Testing, Hypothesis Testing, Time Series Analysis, Data Cleaning, Model Evaluation
- **MLOps:** Model Deployment (FastAPI, Render), Containerization (Docker), Orchestration (Kubernetes), Experiment Tracking (MLFlow), Reproducibility
- **Communication & Collaboration:** Data Storytelling, Stakeholder Communication, Report Writing, Cross-Functional Collaboration, Time Management, Adaptability